

Predicting ribosome P-site density from transcript sequence

dilated convolutional body + transformer, dual-head output

TRAINING DATA

8 human tissues

Cothani Ribo-seq + RNA-seq

Fibroblast

VSMC

ES

Fat

HA_EC

HCAEC

HUVEC

Hepatocytes = HELD OUT

84,472 transcripts (union universe)

validation genes chromosome-disjoint

per-nucleotide input, 10 channels

one-hot seq	4
frame track	5
RNA coverage	1

MODEL

1x1 conv stem

10 -> 256 channels

10 residual dilated blocks

dilation 1

dilation 2

dilation 4

dilation 8

dilation 16

dilation 32

dilation 64

dilation 128

dilation 256

dilation 512

receptive
field ~4 kb

Transformer x2

8 heads, all-to-all

5,071,106 parameters

28 epochs, 14.2 h, 1x A5500

OUTPUT

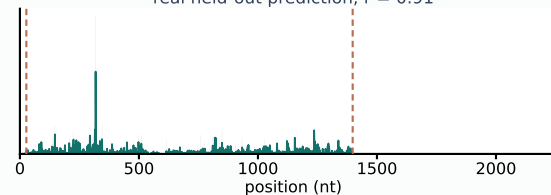
loss = multinomial NLL + 0.1 x MSE

the SHAPE: where translation happens

PROFILE HEAD

softmax over positions -> (B, L)

real held-out prediction, $r = 0.91$



COUNT HEAD

log total P-sites -> (B,)

the DEPTH: how much, dataset-specific

EVALUATION

In-distribution

held-out tissue (Hepatocytes)

profile r 0.659

ORF calling

RiboCode drop-in CDS F1

F1 0.953

Cross-study

Ruiz-Orera (human)

CDS F1 0.931

Cross-species

5 mouse sets (median)

CDS F1 0.971

Proteogenomics

12 macrophage populations

440 novel peptides at 1% class FDR